TOM WILLEMS

Cognitive Neuroscientist & Experimental Psychologist, PhD

tomw92.github.io | tomericwillems@pm.me | +41-76-488-69-59
Stadtbachstrasse 38, 3012 Bern | @tomwillems | tom-willems-8a422a137 | TomW92
0000-0001-9872-0724

EXPERIENCE

Post-Doc

University of Zurich, Psychiatric University Hospital Zurich

03/2026 - present | Zurich, CH

- Experimental Psychopathology and Psychotherapy group led by Prof. Birgit Kleim
- Part of the Clinical Research Priority Program "Synapse, Trauma & Addiction"

Post-Doc

Henke Lab, University of Bern

01/2024 - 12/2025 | Bern, CH

- Focus: applying magnetic resonance spectroscopy (MRS) to investigate how brain metabolites contribute to memory and forgetting

Short Research Stay

Inria Research Centre

03/2024 - 05/2024 | Paris-Saclay, FR

- Models and Inference for Neuroimaging Data (MIND) Group, Prof. Bertrand Thirion
- Focus: across-subject decoding using self-supervised contrastive learning, applied to functional magnetic resonance imaging (fMRI) data

PhD

Henke Lab, University of Bern

10/2018 – 12/2023 | Bern, CH

- Grade: insigni cum laude
- Cognitive Neuroscience of Memory and Consciousness
- Focus: memory, natural forgetting, accessibility of memories

Master of Science: Neuroscience

University of Bonn

10/2015 – 02/2018 | Bonn, DE

- Grade: insigni cum laude
- Focus: cognitive and clinical neuroscience
- Master Thesis: Effects of social anxiety traits on social decision-making: a behavioural and fMRI-study

RESEARCH

- Right now, I am working as a postdoctoral researcher on a randomised placebo-controlled trial examining minocycline's effect on intrusive memory reconsolidation in PTSD and Cocaine Use Disorder.
- During my PhD, I used ultra high-field (7T) fMRI to investigate human memory and forgetting (Piredda et al., 2023; Willems & Henke, 2021). Our findings suggest that early forgetting may reflect a failure of conscious access rather than a loss of the memory itself (Willems et al., 2026).
- For my master thesis, I worked on two fMRI studies investigating social avoidance and guilt (Gädeke et al., 2025; Schultz et al., 2019).
- I am a member of the Swiss Reproducibility Network (SwissRN) Academy, with an active interest in promoting open and reproducible research practices.

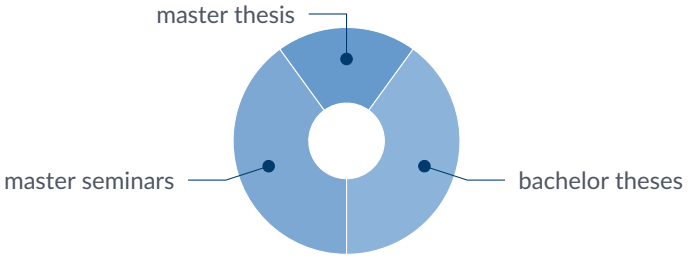
METHODS AND SKILLS

- Multivariate MRI data analysis (functional and anatomical) using Nilearn, SPM, and freesurfer
- Computational skills: advanced proficiency in Python, R, Matlab, and Git
- MRS data analysis
- Computational Modeling in R using 'lme4' and 'marginaleffects' packages
- Scientific writing
- Study design in experimental psychology

GRANTS AND AWARDS

- UniBE Short Travel Grants for (Post)Docs, 3'210 CHF (2024)
- Swiss Dementia Forum - Poster Award, Swiss Network for Dementia Research, 500 CHF (2025)

TEACHING / MENTORING



Teaching:

- 5 seminars for master level psychology students, on topics such as the hippocampal role in memory processing, neurogenesis in the adult hippocampus, and language and semantics.

Mentoring:

- 4 bachelor thesis
- 1 master thesis

REFERENCES

Prof. Katharina Henke
@ Institute of Psychology
katharina.henke@unibe.ch
University of Bern, Fabrikstrasse 8, 3012 Bern, Switzerland

Prof. Bertrand Thirion
@ Head of science (délégué scientifique) Inria Saclay
bertrand.thirion@inria.fr
Neurospin, CEA Saclay, 91191 Gif sur Yvette, France

PD Dr. Johannes Schultz
@ Acting Director - Center for Economics and Neuroscience
johannesschultz@uni-bonn.de
Universität Bonn, Am Hofgarten 8, 53113 Bonn, Germany

LANGUAGES

German
native

English
full professional
work proficiency

French
limited work proficiency

PUBLICATIONS

- Willems, T., Zervas, K., Brogli, L., Rabe, F., Federspiel, A., & Henke, K. (2026). Neural traces of forgotten memories persist in humans and are behaviorally relevant. *eLife*. doi:10.7554/elife.109530.1
- Gädeke, M., Willems, T., Ahmed, O. S., Weber, B., Hurlemann, R., & Schultz, J. (2025). Contributions of insula and superior temporal sulcus to interpersonal guilt and responsibility in social decisions. *eLife*. doi:10.7554/elife.105391.1
- Piredda, G. F., Caneschi, S., Hilbert, T., Bonanno, G., Joseph, A., Egger, K., … Radojewski, P. (2023). Submillimeter T1 atlas for subject-specific abnormality detection at 7T. *Magnetic Resonance in Med*, *89*(4), 1601–1616. doi:10.1002/mrm.29540
- Willems, T., & Henke, K. (2021). Imaging human engrams using 7 Tesla magnetic resonance imaging. *Hippocampus*, *31*(12), 1257–1270. doi:10.1002/hipo.23391
- Schultz, J., Willems, T., Gädeke, M., Chakkour, G., Franke, A., Weber, B., & Hurlemann, R. (2019). A human subcortical network underlying social avoidance revealed by risky economic choices. *eLife*, *8*, e45249. doi:10.7554/eLife.45249